

ESP32 WATER QUALITY MONITORING SYSTEM - SCHEMATIC

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Rev: 1.0

Date: 2024-01-15

POWER SUPPLY

BAT1: 7.4V LiPo Input
F1: PTC Fuse 2A
U3: AMS1117-3.3 (3.3V/1A)
U4: AMS1117-5.0 (5.0V/1A)
C9,C10: 1000uF filter caps
D1: 1N5819 reverse protection

ESP32-WROOM-32

U1: ESP32 Main Controller
GPIO4: LoRa CS
GPIO5: LoRa SCK
GPIO18: LoRa MISO
GPIO19: LoRa MOSI
GPIO16: GPS RX
GPIO17: GPS TX

SENSOR INTERFACE

U7: MCP3008 ADC (SPI)
CH0: pH Sensor
CH1: Turbidity Sensor
GPIO15: 1-Wire (DS18B20)
R13: 4.7K pullup

MOTOR DRIVER

U2: L298N Dual H-Bridge
IN1: GPIO25
IN2: GPIO26
ENA: GPIO27 (PWM)
OUT1,OUT2: Pump Motor

LORA TELEMETRY

U5: RFM95W 915MHz
NSS: GPIO4
SCK: GPIO5
MISO: GPIO18
MOSI: GPIO19
DIO0: GPIO2

GPS MODULE

U6: NEO-6M GPS
TX: GPIO16 (RX2)
RX: GPIO17 (TX2)
Baud: 9600
Update: 10Hz

DATA LOGGING

J5: MicroSD Socket
CS: GPIO13
SCK: GPIO14
MISO: GPIO12
MOSI: GPIO23
Format: FAT32

STATUS INDICATORS

D2: Power LED (Red)
D3: GPS Lock (Green)
D4: LoRa TX (Blue)
D5: Sampling (Red)
D6: Error (Red)
BZ1: Piezo Buzzer

EXTERNAL SENSORS

J1: BNC (pH Sensor)
J2: 4-Pin (Turbidity)
J3: 2-Pin (Pump Motor)
J4: 3-Pin (DS18B20)
X1: Temperature Sensor

ESP32 GPIO ASSIGNMENTS

GPIO2: LoRa DIO0 | GPIO4: LoRa CS | GPIO5: LoRa SCK | GPIO12: SD MISO | GPIO13: SD CS
GPIO14: SD SCK | GPIO15: 1-Wire | GPIO16: GPS RX | GPIO17: GPS TX | GPIO18: LoRa MISO
GPIO19: LoRa MOSI | GPIO21: I2C SDA | GPIO22: I2C SCL | GPIO23: SD MOSI | GPIO25: Motor IN1
GPIO26: Motor IN2 | GPIO27: Motor PWM | GPIO32: ADC CS | GPIO33: ADC SCK | GPIO34: LED Status

DESIGN NOTES:

- All decoupling capacitors should be placed close to IC power pins
- LoRa and GPS modules require external antennas with proper impedance matching
- Keep high-frequency traces (SPI, I2C) short and direct
- Use star grounding topology for analog sensor ground

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